Worked Solutions: Powers & Exponents (10.1–10.4) — Worksheet #1

Section A: Simplify (Problems 1–4)

Recall the **Product of Powers Rule**: $a^m \cdot a^n = a^{m+n}$, and the **Power of a Power Rule**: $(a^m)^n = a^{mn}$.

1. Original expression: $x^3 \cdot x^5$

$$\begin{aligned} x^3 \cdot x^5 &= x^{3+5} && \text{(Product of powers: add exponents)} \\ &= x^8 \end{aligned}$$

$$\boxed{x^8}$$

2. Original expression: $(y^4)^3$

$$\begin{aligned} (y^4)^3 &= y^{4\cdot 3} && \text{(Power of a power: multiply exponents)} \\ &= y^{12} \end{aligned}$$

$$\boxed{y^{12}}$$

3. Original expression: $a^7 \cdot a^{-2}$

$$\begin{aligned} a^7 \cdot a^{-2} &= a^{7+(-2)} && \text{(Product of powers)} \\ &= a^5 \end{aligned}$$

$$\boxed{a^5}$$

4. Original expression: $(b^2)^0$

$$\begin{aligned} (b^2)^0 &= b^{2\cdot 0} && \text{(Power of a power, and any base to zero is 1)} \\ &= b^0 \\ &= 1 \end{aligned}$$

$$\boxed{1}$$

Section B: Evaluate (Problems 5–8)

Here we expand powers and multiply or divide numbers step-by-step.

5. Original expression: $2^3 \cdot 3^2$

$$\begin{aligned} 2^3 \cdot 3^2 &= (2 \cdot 2 \cdot 2) \cdot (3 \cdot 3) \\ &= 8 \cdot 9 \\ &= 72 \end{aligned}$$

$$\boxed{72}$$

6. Original expression: $\frac{5^4}{5^2}$

$$\begin{aligned} \frac{5^4}{5^2} &= 5^{4-2} && \text{(Quotient of powers: subtract exponents)} \\ &= 5^2 \\ &= 25 \end{aligned}$$

$$\boxed{25}$$

7. Original expression: $(4^2)^3$

$$\begin{aligned} (4^2)^3 &= 4^{2\cdot 3} && \text{(Power of a power)} \\ &= 4^6 \\ &= 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \cdot 4 \\ &= 4096 \end{aligned}$$

$$\boxed{4096}$$

8. Original expression: $\frac{10^5}{2 \cdot 5^3}$

$$\begin{aligned} \frac{10^5}{2 \cdot 5^3} &= \frac{(2 \cdot 5)^5}{2 \cdot 5^3} && \text{(Write } 10 = 2 \cdot 5\text{)} \\ &= \frac{2^5 \cdot 5^5}{2 \cdot 5^3} \\ &= \frac{2^5}{2^1} \cdot \frac{5^5}{5^3} && \text{(Separate powers)} \\ &= 2^{5-1} \cdot 5^{5-3} && \text{(Quotient of powers)} \\ &= 2^4 \cdot 5^2 \\ &= 16 \cdot 25 \\ &= 400 \end{aligned}$$

$$\boxed{400}$$

Section C: Simplify Completely (Problems 9–13)

We will use exponent rules including product, quotient, power of a power, and negative exponent rules. Recall the **Negative Exponent Rule**: $a^{-n} = \frac{1}{a^n}$.

9. Original expression: $\frac{x^7}{x^3}$

$$\begin{aligned}\frac{x^7}{x^3} &= x^{7-3} && \text{(Quotient of powers)}\\ &= x^4\end{aligned}$$

$$\boxed{x^4}$$

10. Original expression: $(m^3n^2)^4$

$$\begin{aligned}(m^3n^2)^4 &= m^{3\cdot 4}n^{2\cdot 4} && \text{(Power of a product)}\\ &= m^{12}n^8\end{aligned}$$

$$\boxed{m^{12}n^8}$$

11. Original expression: $\frac{a^5b^{-2}}{a^{-3}b^4}$

$$\begin{aligned}\frac{a^5b^{-2}}{a^{-3}b^4} &= a^{5-(-3)}b^{-2-4} && \text{(Quotient of powers)}\\ &= a^{5+3}b^{-6}\\ &= a^8b^{-6}\end{aligned}$$

$$\boxed{a^8b^{-6}}$$

12. Original expression: $(2x^3y^{-1})^2$

$$\begin{aligned}(2x^3y^{-1})^2 &= 2^2\cdot(x^3)^2\cdot(y^{-1})^2\\ &= 4\cdot x^{3\cdot 2}\cdot y^{-2}\\ &= 4x^6y^{-2}\end{aligned}$$

$$\boxed{4x^6y^{-2}}$$

13. Original expression: $\frac{(p^2q^{-3})^3}{p^{-4}q^2}$

$$\begin{aligned}\frac{(p^2q^{-3})^3}{p^{-4}q^2} &= \frac{p^{2\cdot 3}q^{-3\cdot 3}}{p^{-4}q^2}\\ &= \frac{p^6q^{-9}}{p^{-4}q^2}\\ &= p^{6-(-4)}q^{-9-2} && \text{(Quotient of powers)}\\ &= p^{10}q^{-11}\end{aligned}$$

$$\boxed{p^{10}q^{-11}}$$

Section D: Simplify Completely (Problems 14–17)

Again, apply all exponent rules carefully.

14. Original expression: $\left(\frac{x^2y^{-3}}{x^{-1}y^4}\right)^2$

$$\begin{aligned}
\left(\frac{x^2y^{-3}}{x^{-1}y^4}\right)^2 &= \left(x^{2-(-1)}y^{-3-4}\right)^2 && \text{(Quotient of powers)}\\
&= \left(x^3y^{-7}\right)^2\\
&= x^{3\cdot 2}y^{-7\cdot 2} && \text{(Power of a power)}\\
&= x^6y^{-14}
\end{aligned}$$

$\boxed{x^6y^{-14}}$

15. Original expression: $\frac{(2a^{-1}b^3)^2}{4a^2b^{-2}}$

$$\begin{aligned}
\frac{(2a^{-1}b^3)^2}{4a^2b^{-2}} &= \frac{2^2\cdot a^{-1\cdot 2}\cdot b^{3\cdot 2}}{4a^2b^{-2}}\\
&= \frac{4a^{-2}b^6}{4a^2b^{-2}}\\
&= \frac{4}{4}\cdot a^{-2-2}\cdot b^{6-(-2)} && \text{(Quotient of powers)}\\
&= a^{-4}b^8
\end{aligned}$$

$\boxed{a^{-4}b^8}$

16. Original expression: $(x^0y^5)^{-3}$

$$\begin{aligned}
(x^0y^5)^{-3} &= x^{0\cdot(-3)}y^{5\cdot(-3)} && \text{(Power of a product)}\\
&= x^0y^{-15}\\
&= y^{-15}
\end{aligned}$$

$\boxed{y^{-15}}$

17. Original expression: $x^{-2}\cdot\frac{1}{x^{-5}}$

$$\begin{aligned}
x^{-2}\cdot\frac{1}{x^{-5}} &= x^{-2}\cdot x^5 && \text{(Negative exponent rule)}\\
&= x^{-2+5} && \text{(Product of powers)}\\
&= x^3
\end{aligned}$$

$\boxed{x^3}$

Section E: Simplify with Positive Exponents (Problems 18–22)

Write final answers with only positive exponents.

18. Original expression: $\frac{x^{-3}y^4}{x^2y^{-1}}$

$$
\begin{aligned}
\frac{x^{-3}y^4}{x^2y^{-1}} &= x^{-3-2}y^{4-(-1)} && \text{(Quotient of powers)}\\
&= x^{-5}y^5\\
&= \frac{y^5}{x^5} && \text{(Negative exponent rule)}
\end{aligned}
$$

$$\boxed{\frac{y^5}{x^5}}$$

19. Original expression: $(a^{-2}b^3)^2$

$$
\begin{aligned}
(a^{-2}b^3)^2 &= a^{-2\cdot 2}b^{3\cdot 2} && \text{(Power of a product)}\\
&= a^{-4}b^6\\
&= \frac{b^6}{a^4} && \text{(Negative exponent rule)}
\end{aligned}
$$

$$\boxed{\frac{b^6}{a^4}}$$

20. Original expression: $\frac{(x^3y^{-2})^{-2}}{x^{-4}y^3}$

$$
\begin{aligned}
\frac{(x^3y^{-2})^{-2}}{x^{-4}y^3} &= \frac{x^{3\cdot(-2)}y^{-2\cdot(-2)}}{x^{-4}y^3}\\
&= \frac{x^{-6}y^4}{x^{-4}y^3}\\
&= x^{-6-(-4)}y^{4-3} && \text{(Quotient of powers)}\\
&= x^{-2}y^1\\
&= \frac{y}{x^2} && \text{(Negative exponent rule)}
\end{aligned}
$$

$$\boxed{\frac{y}{x^2}}$$

21. Original expression: $\left(\frac{a^{-1}b^2}{a^3b^{-4}}\right)^{-2}$

$$
\begin{aligned}
\left(\frac{a^{-1}b^2}{a^3b^{-4}}\right)^{-2} &= \left(a^{-1-3}b^{2-(-4)}\right)^{-2} && \text{(Quotient of powers)}\\
&= \left(a^{-4}b^6\right)^{-2}\\
&= a^{-4\cdot(-2)}b^{6\cdot(-2)} && \text{(Power of a power)}\\
&= a^8b^{-12}\\
&= \frac{a^8}{b^{12}} && \text{(Negative exponent rule)}
\end{aligned}
$$

$$\boxed{\frac{a^8}{b^{12}}}$$

22. Original expression: $\frac{(x^{-2}y^3)^3}{(xy^{-1})^4}$

$$\begin{aligned}
\frac{(x^{-2}y^3)^3}{(xy^{-1})^4} &= \frac{x^{-2\cdot 3}y^{3\cdot 3}}{x^4y^{-1\cdot 4}} \\
&= \frac{x^{-6}y^9}{x^4y^{-4}} \\
&= x^{-6-4}y^{9-(-4)} && \text{(Quotient of powers)} \\
&= x^{-10}y^{13} \\
&= \frac{y^{13}}{x^{10}} && \text{(Negative exponent rule)}
\end{aligned}$$

$$\boxed{\frac{y^{13}}{x^{10}}}$$

Section F: Word Problems (Problems 23–25)

Show unit conversions and exponent arithmetic carefully. Final answers for problem 25 must be in standard form.

23. Original problem: A cube has side length 5×10^2 cm. Find its volume in cubic meters.

$$\begin{aligned}
\text{Side length} &= 5 \times 10^2 \text{ cm} \\
&= 5 \times 10^2 \times \frac{1 \text{ m}}{100 \text{ cm}} && \text{(Convert cm to m)} \\
&= 5 \times 10^2 \times 10^{-2} \text{ m} \\
&= 5 \times 10^{2-2} \text{ m} = 5 \times 10^0 \text{ m} = 5 \text{ m} \\
\text{Volume} &= (\text{side length})^3 = (5 \text{ m})^3 = 5^3 \text{ m}^3 = 125 \text{ m}^3
\end{aligned}$$

$$\boxed{125 \text{ m}^3}$$

24. Original problem: The population of a city is 3×10^6. If it doubles every 5 years, what is the population after 10 years?

$$\begin{aligned}
\text{After 5 years:} &\quad 3 \times 10^6 \times 2 = 6 \times 10^6 \\
\text{After 10 years:} &\quad 6 \times 10^6 \times 2 = 12 \times 10^6 = 1.2 \times 10^7
\end{aligned}$$

$$\boxed{1.2 \times 10^7}$$

25. Original problem: Express $(4 \times 10^{-3})^3$ in standard form.

$$\begin{aligned}
(4 \times 10^{-3})^3 &= 4^3 \times (10^{-3})^3 && \text{(Power of a product)} \\
&= 64 \times 10^{-9} \\
&= 6.4 \times 10^1 \times 10^{-9} && \text{(Rewrite } 64 = 6.4 \times 10^1\text{)} \\
&= 6.4 \times 10^{1-9} \\
&= 6.4 \times 10^{-8}
\end{aligned}$$

$$\boxed{6.4 \times 10^{-8}}$$